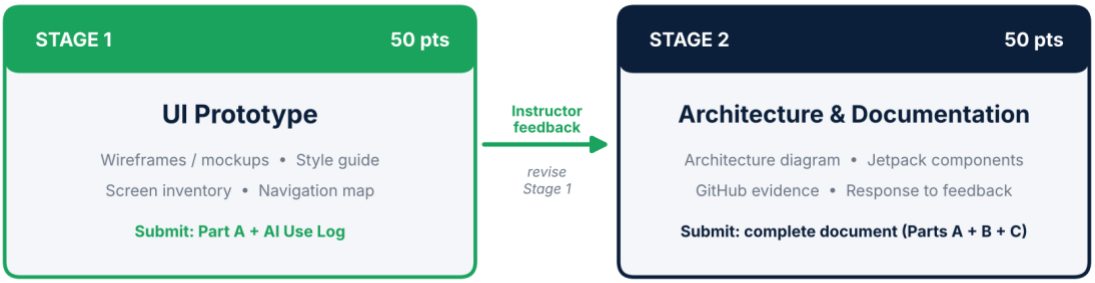
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CSC 402 — MILESTONE 2

NEXA — Milestone 2

UI/UX Design & System Architecture • Two-stage submission



How this milestone works

Stage	Submit	Due	Points	What you hand in
Stage 1	UI Prototype	Week 6	50	Cover sheet, Part A (app recap, style guide, wireframes, screen inventory, navigation map) and Part C (AI Use Log entries so far)
Stage 2	Complete Milestone 2	Week 7	50	This whole document: Part A revised after feedback, plus Part B (architecture, Jetpack components, GitHub evidence) and the final Part C

How you will be graded

Milestone 2 is scored out of 100 points using the Milestone 2 rubric. The points are split between the two stages as follows.

Stage	Rubric criterion	Points	Where in this template
Stage 1	Wireframes / Mockups	25	A3 UI Prototype
Stage 1	Navigation Map	15	A4 Navigation Map
Stage 1	Design Consistency & Professionalism	10	A2 Design Foundations and all figures
Stage 2	Architecture Diagram	25	B2 System Architecture
Stage 2	Jetpack Component Plan	15	B3 Jetpack Components
Stage 2	GitHub Documentation	10	A5 and B4 GitHub Evidence

AI-related disclosure is checked through the AI Use Log in Part C. It is graded for honesty and clarity only, not for whether you used AI.

Submission Cover Sheet

Course / Milestone	CSC 402 Mobile Application Programming — Milestone 2

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Team name	Team 05
App name	NEXA — "One place. Every project."
GitHub repository	https://github.com/Hebaasksr/Nexa.git
Stage 1 tag / commit	milestone-2-stage-1 / ef9786960680284369dd35a384fe3a9688013fd0
Stage 2 tag / commit	milestone-2 / c7746a761726557048058b700f18084fab658b9a
Prototype link	https://www.figma.com/design/kHE5HexxH6KgkDjKnfB4qH/NEXA-—-HCI-7-Mobile-Interfaces?node-id=25-34337&t=Hq2BBInH2KUE0MZp-1
Submission dates	Stage 1: 6 October 2026 Stage 2: 26 October 2026

Team roster and contributions

#	Member name	Student ID	Role	Stage 1 contribution	Stage 2 contribution
1	**Ghaida Hashem AlQarni**	2240004359	Problem & Target Users	Developed the problem statement and analyzed the target users and their needs, and contributed to designing several NEXA UI/UX wireframes.	Contributed to designing NEXA UI/UX wireframes. Uploaded the wireframes, navigation map, style guide, and AI Use Log to GitHub, and updated the README.md file to organize and document the project's design resources.
2	**Fatimah Emad Alarfaj**	2240007580	Proposal Documentation	I contributed to selecting the libraries used in the project and provided the reasons and justification for choosing them.	Documented the Jetpack Components and Libraries for NEXA Milestone 2, explaining the purpose of each component and justifying its selection based on the application's requirements and system architecture. Created and merged Pull Request #11 to add the documentation to the GitHub repository
3	**Heba Khaled Askar**	2240006064	Project Manager & Coordinator	I contributed to designing and building the user interface using Figma, focusing on the layout, visual design, and overall user experience.	Coordinated Stage 2 activities, reviewed the submission for consistency and completeness, and organized the final milestone documentation according to the submission requirements.

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#	Member name	Student ID	Role	Stage 1 contribution	Stage 2 contribution
4	**Munirah Hassan Albuainain**	2240007600	MVP Features & Risks	Defined and prioritized the application's core features, and designed the System Architecture diagram with its Unidirectional Data Flow.	Designed the System Architecture diagram and documented the Unidirectional Data Flow, explicitly defining the boundaries and interactions between the UI, Domain, and Data layers.
5	**Jawaher Shaya Hazazi**	2240006281	Proposed Solution & Project Value	Designed the application's Navigation Map, defining the main user flow and navigation between the Login, Dashboard, Projects, Tasks, and Calendar sections to ensure a clear and organized user experience.	Updated the navigation map to include missing screens and navigation flows, aligned screen names with the UI prototype, and clarified project, task, and team navigation paths.

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PART A STAGE 1 — UI Prototype

Submit in Stage 1 and again (revised after feedback) in Stage 2 • 50 points

A1 App Recap (from Milestone 1)

Problem	University students coordinate collaborative academic projects across several scattered messaging platforms, which causes high cognitive load, miscommunication, and unclear task dependencies. There is no centralized, end-to-end workflow tracking system that gives real-time insight into group dynamics and project progress.
Target users	University students who frequently work on collaborative academic projects and need one intuitive place to organize tasks, manage team responsibilities, and monitor deadlines, instead of juggling multiple disjointed apps.

Feature	Short description
User Registration & Login	Students create an account, securely log in and reset a forgotten password. (Must)
Create & Manage Projects	Create projects with details, dates and team; manage active projects. (Must)
Join Project / Team	Join a project using a project code or an invitation. (Must)
Task Management	Create, assign, edit and complete tasks. (Must)
Task Deadlines	Set deadlines for tasks and view upcoming deadlines. (Must)
Project Progress Tracking	See overall project progress based on completed and remaining tasks. (Must)
Team Members	View team members and their assigned responsibilities. (Must)
Dashboard	Overview of active projects, pending and completed tasks, and upcoming deadlines. (Must)
File Sharing	Upload and access files related to a project. (Should)
Notifications & Reminders	Receive reminders about upcoming deadlines and task updates. (Should)
Project Calendar	View project deadlines and important dates in a calendar. (Should)
Task Comments	Comment on tasks to discuss updates or requirements. (Should)
Profile Management	View and update profile information and preferences. (Could)
Search & Filter Tasks	Search and filter tasks by status, deadline, or assigned member. (Could)
Project Activity Log	Show recent project activities such as completed tasks, new files, and assignments. (Could)

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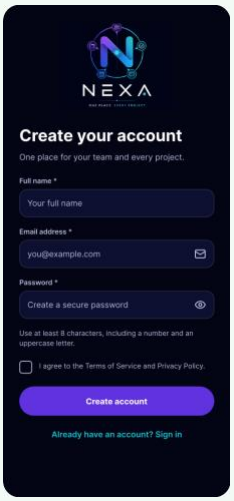
A2 Design Foundations (Style Guide)

Element	Your decision	Used for
Primary color	Electric Purple ≈ #6133E0 on deep navy (≈ #040514) dark surfaces	Primary buttons (Sign In, Create account, Create task), the centre "New" button, active tabs, progress bars, links to detail screens.
Accent color	Cyan / Teal ≈ #11A7B6	Secondary links ("Forgot password?", "Sign in"), card edge highlights, "In Progress" and calendar markers.
Status colors	Green = success (Active, Completed, Online); Amber = warning (Due today, medium priority, Away); Red ≈ #B02038 = error / destructive	Status badges, priority tags, validation errors, and the "Delete task" confirmation.
Typography	Clean sans-serif (Inter-style) with Material 3 type scale: Headline, Title, Body, Label	Headline: Screen titles. Title: section and card headings. Body: descriptions and inputs. Label: badges, tabs, and navigation labels.
Spacing and shape	Dark gradient header with light rounded content sheet; rounded cards; pill-shaped buttons and tabs; 8 dp spacing grid	Screen layout, cards (projects, tasks, members, notifications) and buttons.
Components	Material 3: Button / OutlinedButton, OutlinedTextField, Card, NavigationBar, TopAppBar, TabRow, LinearProgressIndicator, AlertDialog, Switch, Checkbox, dropdown menu, date picker	Auth forms, bottom navigation (Home, Tasks, New, Calendar, More), My Tasks filters, project progress, delete dialog, settings toggles.

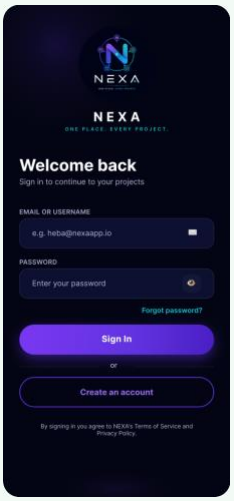
A3 UI Prototype

Design tool	Figma
Prototype link	https://www.figma.com/design/kHE5HexxH6KgkDjKnfB4qH/NEXA-—-HCI-7-Mobile-Interfaces?node-id=25-34337&t=Hq2BBInH2KUE0MZp-1
Number of screens	21 wireframes

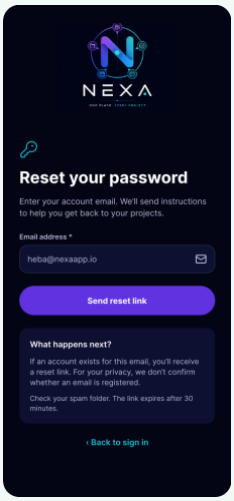
Screen 1
Create Account



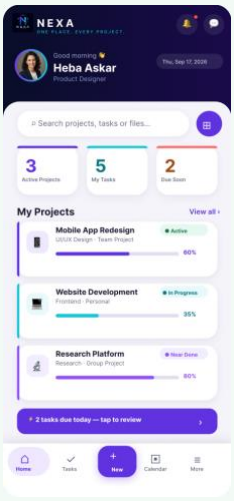
Screen 2
Sign In



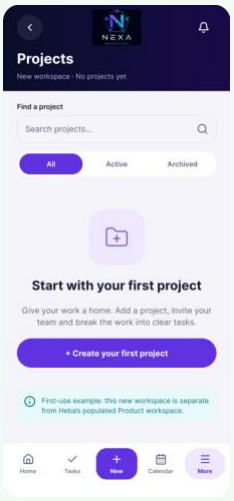
Screen 3
Reset Password



Screen 4
Dashboard (Home)

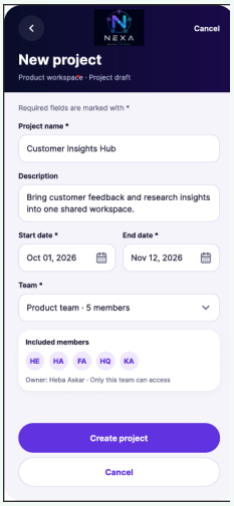


Screen 5
Projects (empty state)



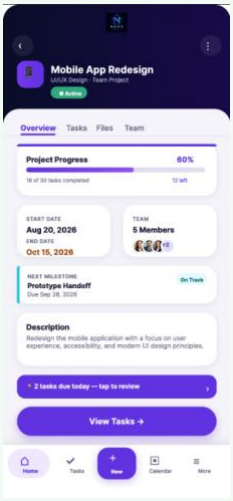
Screen 6

New Project



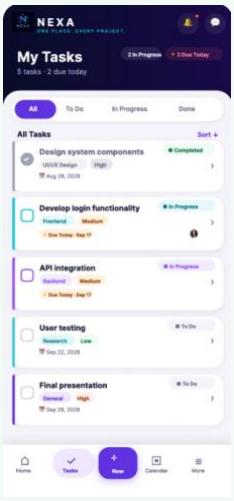
Screen 7

Project Details



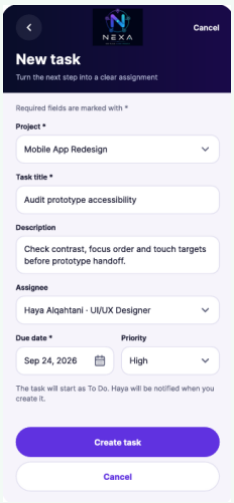
Screen 8

My Tasks



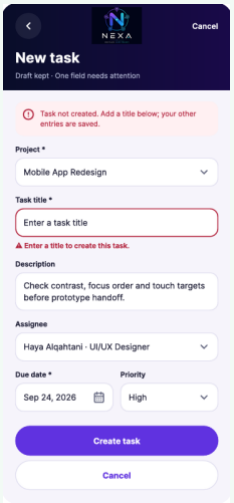
Screen 9

New Task



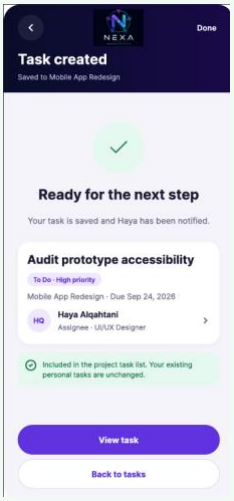
Screen 10

New Task — validation error



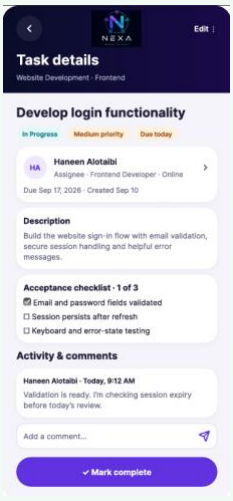
Screen 11

Task Created



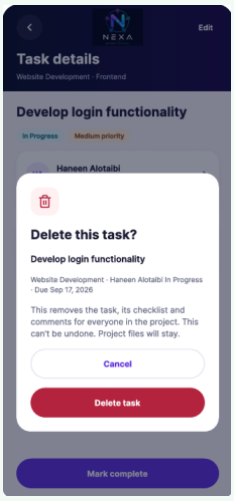
Screen 12

Task Details



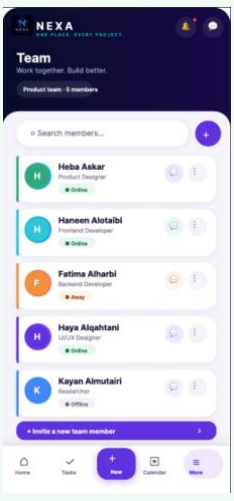
Screen 13

Delete Task (dialog)



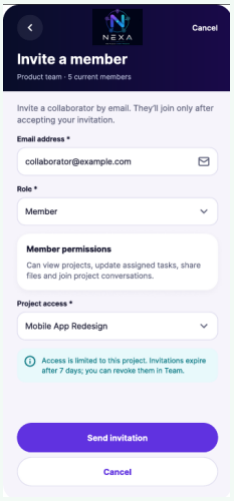
Screen 14

Team



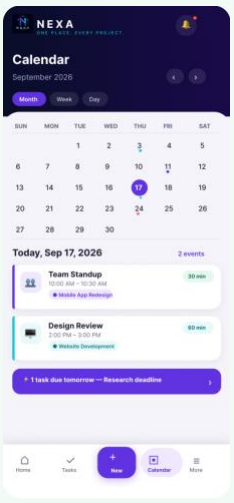
Screen 15

Invite a Member



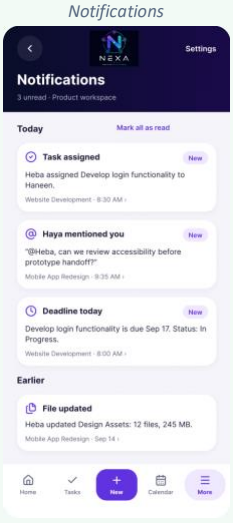
Screen 16

Calendar



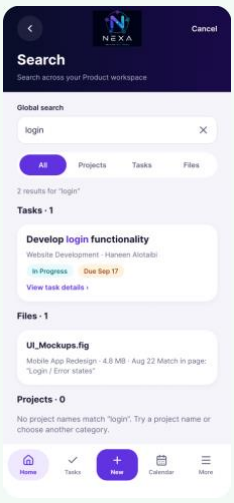
Screen 17

Notifications



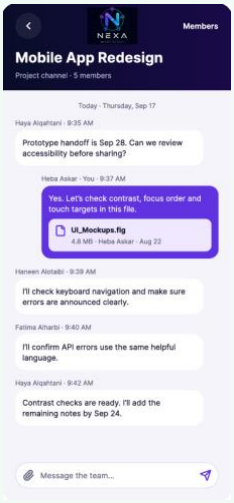
Screen 18

Search



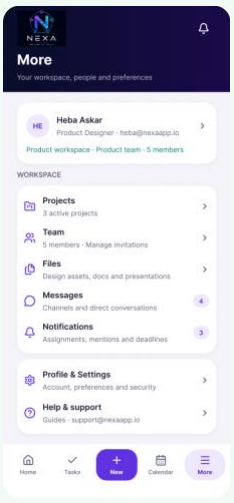
Screen 19

Project Chat



Screen 20

More



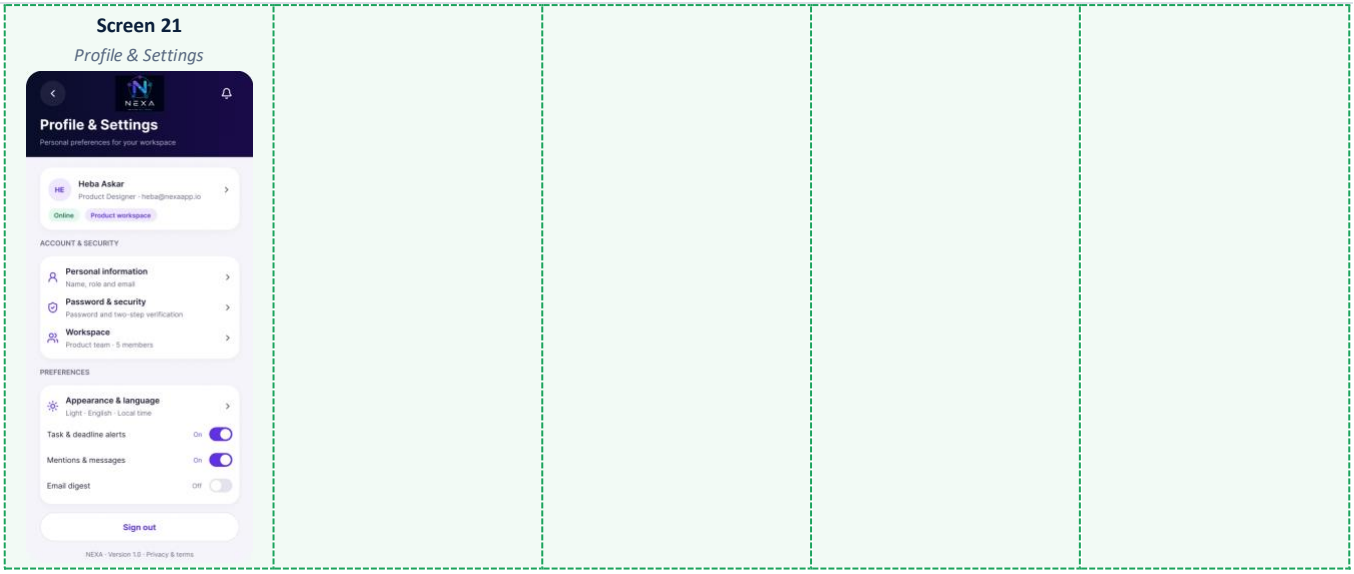


Figure 1 — NEXA prototype screens: Create Account, Sign In, Reset Password and Dashboard (1–4); Projects, New Project, Project Details (5–7); My Tasks, New Task, validation error, Task Created, Task Details, Delete Task dialog (8–13); Team, Invite a Member (14–15); Calendar, Notifications, Search, Project Chat (16–19); More and Profile & Settings (20–21).

Screen inventory

#	Screen	Purpose	Key elements	Feature
1	**Create Account**	Register a new student account	Full name, email, password with rules, Terms checkbox, "Create account", link to sign in	User Registration & Login
2	**Sign In**	Log in to reach projects	Email/username and password fields, "Forgot password?", "Sign In", "Create an account"	User Registration & Login
3	**Reset Password**	Recover account access	Email field, "Send reset link", "What happens next?" card, back to sign in	User Registration & Login
4	**Dashboard (Home)**	Overview of work and deadlines	Greeting, search bar, counters (Active Projects, My Tasks, Due Soon), My Projects cards with progress and status, "tasks due today" banner, bottom navigation	Dashboard; Project Progress Tracking
5	**Projects (empty state)**	Find and list projects; first-use state	Search field, All / Active / Archived tabs, empty-state message, "Create your first project"	Create & Manage Projects
6	**New Project**	Create a project	Name, description, start and end dates, team, included members, "Create project", Cancel	Create & Manage Projects
7	**Project Details**	See one project at a glance	Overview / Tasks / Files / Team tabs, progress bar (60%), dates, team avatars, next milestone, description, "View Tasks"	Project Progress Tracking; Team Members
8	**My Tasks**	Browse and track tasks	All / To Do / In Progress / Done filter, task cards with tags, priority, due date and checkbox, Sort	Task Management; Task Deadlines; Search & Filter Tasks
9	**New Task**	Create and assign a task	Project, title, description, assignee, due date, priority, "Create task", Cancel	Task Management; Task Deadlines
10	**New Task — validation error**	Show a clear error without losing input	Error banner, highlighted title field with inline message, other entries kept	Task Management

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#	Screen	Purpose	Key elements	Feature
11	**Task Created**	Confirm the task was saved	Success check, task summary (priority, due date, assignee), "View task", "Back to tasks"	Task Management
12	**Task Details**	View and update one task	Status, priority and due badges, assignee, description, acceptance checklist, activity & comments with comment box, "Mark complete", Edit	Task Management; Task Comments
13	**Delete Task (dialog)**	Confirm a destructive action	Warning text, "Cancel", "Delete task"	Task Management
14	**Team**	See team members and responsibilities	Search members, member cards (role, online status, chat button), "+" and "Invite a new team member"	Team Members
15	**Invite a Member**	Invite a collaborator by email	Email, role, member permissions, project access, invitation note, "Send invitation", Cancel	Join Project / Team
16	**Calendar**	See deadlines and events by date	Month / Week / Day toggle, month grid with markers, selected-day events, "task due tomorrow" banner	Project Calendar
17	**Notifications**	Review reminders and updates	Today / Earlier groups, Task assigned, mention, Deadline today and File updated cards, "Mark all as read", Settings	Notifications & Reminders
18	**Search**	Search across projects, tasks and files	Search field, All / Projects / Tasks / Files filters, grouped results, "no results" message	Search & Filter Tasks
19	**Project Chat**	Discuss a project with the team	Message bubbles, shared file chip, message box with attach and send, Members	Task Comments; File Sharing
20	**More**	Hub for workspace and preferences	Profile card, Projects, Team, Files, Messages, Notifications, Profile & Settings, Help & support	Profile Management
21	**Profile & Settings**	View and update profile and preferences	Profile card, personal information, password & security, workspace, appearance, alert toggles, "Sign out"	Profile Management

UI states

State	Designed?	Screen(s)
Loading	☐ Yes ☒ No	Not designed yet
Empty list / no data	☒ Yes ☐ No	Projects (empty state: "Start with your first project"); Search ("Projects · 0" no-results message)
Error / no network	☒ Yes ☐ No	New Task — validation error ("Task not created", inline title error). A no-network state is not designed yet and is a Milestone 3 issue.
Selected / active element	☒ Yes ☐ No	Bottom navigation (active tab), Projects tabs, My Tasks filter, Calendar selected date, Search filter
Destructive action confirmation	☒ Yes ☐ No	Delete Task dialog ("Delete this task?")

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A4 Navigation Map

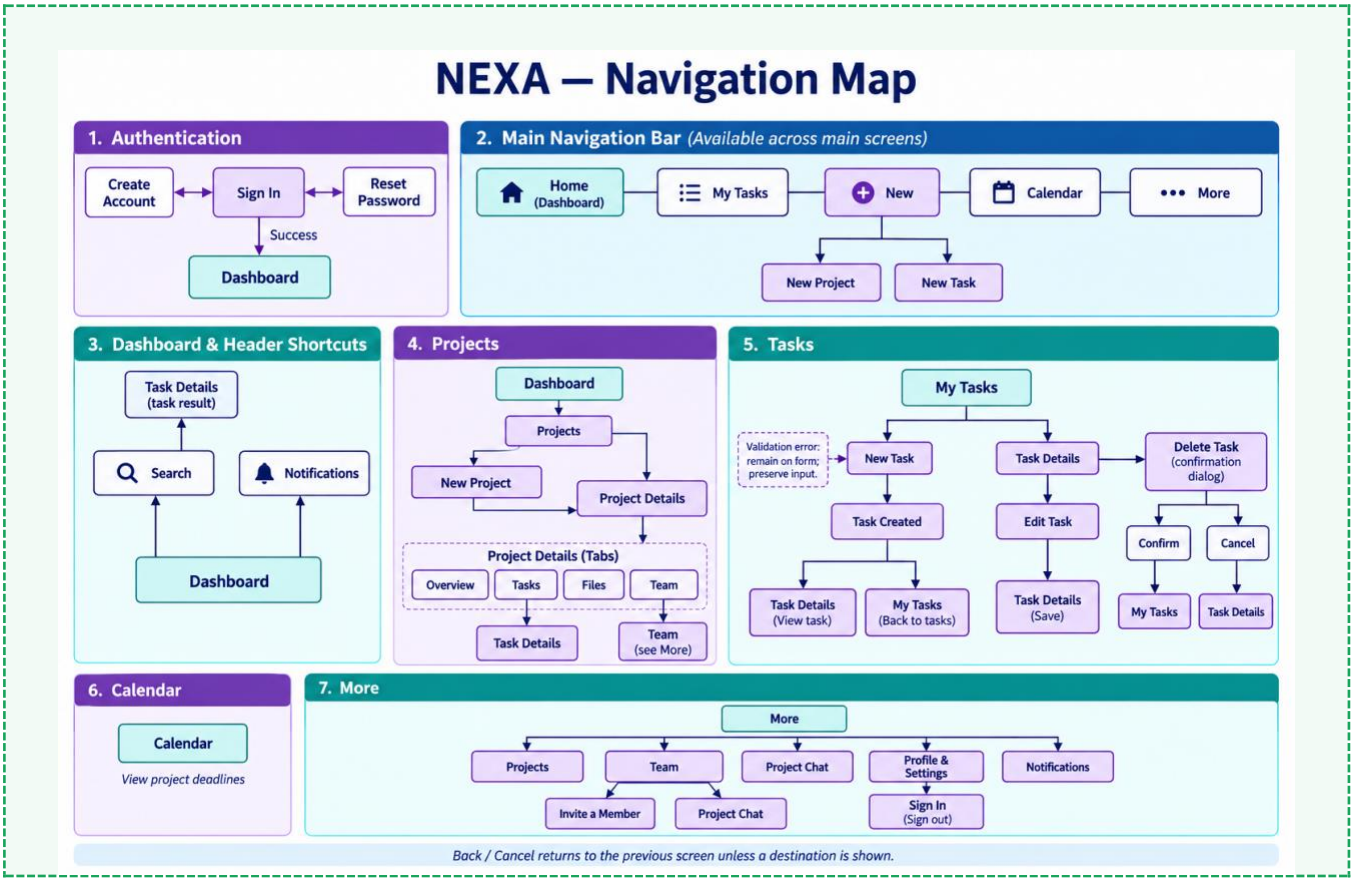


Figure 2 — NEXA navigation map: Sign In with Create Account and Reset Password; main navigation through Dashboard, My Tasks, New, Calendar, and More; project creation and Project Details tabs (Overview, Tasks, Files, Team); task creation, viewing, editing, and deletion confirmation; team invitations, project chat, search, notifications, and profile settings.

Route	Screen	Arguments	Reached from	Back behavior
register	Create Account	—	Sign In → "Create an account"	Back to login
login	Sign In	—	App start (signed out)	Exits app; after success navigates to dashboard and clears login
reset_password	Reset Password	—	Sign In → "Forgot password?"	Back to login
dashboard	Dashboard (Home) — tab	—	Login; bottom nav "Home"	Exits app (start of tab stack)
tasks	My Tasks — tab	status filter (optional)	Bottom nav "Tasks"; Dashboard banner	Returns to dashboard
calendar	Calendar — tab	—	Bottom nav "Calendar"	Returns to dashboard
more	More — tab	—	Bottom nav "More"	Returns to dashboard
projects	Projects	—	Dashboard "View all"; More → Projects	Previous screen
new_project	New Project	—	Bottom nav "New"; Projects empty state	Cancel / back → Projects

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Route	Screen	Arguments	Reached from	Back behavior
project/{projectId}	Project Details (Overview, Tasks, Files, Team)	projectId	Dashboard project card; Projects list	Back to Projects
new_task	New Task (incl. validation error state)	projectId (optional)	Bottom nav "New"; My Tasks	Cancel / back → previous; success → task_created
task_created/{taskId}	Task Created	taskId	New Task → "Create task"	"Back to tasks" → tasks; "View task" → task/{taskId}
task/{taskId}	Task Details	taskId	My Tasks list; Task Created; Search results	Back to My Tasks
edit_task/{taskId}	Edit Task (see navigation map)	taskId	Task Details → Edit	Back to Task Details
delete_task	Delete Task (dialog)	taskId	Task Details / Edit Task → delete	Cancel closes dialog; confirm → tasks
team	Team	projectId	More → Team; Project Details → Team tab	Previous screen
invite_member	Invite a Member	projectId	Team → "+" / "Invite a new team member"	Cancel / back → Team
notifications	Notifications	—	Bell icon in headers; More → Notifications	Previous screen
search	Search	query (optional)	Dashboard search bar	Cancel / back → previous
chat/{projectId}	Project Chat	projectId	More → Messages; Team chat button	Previous screen
profile	Profile & Settings	—	More → profile card / Profile & Settings	Back to More

A5 Stage 1 GitHub Evidence

- ☑ Wireframe/mockup images for every screen are in /docs/wireframes
- ☑ Navigation map image is in /docs/navigation-map.png
- ☑ docs/style-guide.md is committed
- ☑ README links to the prototype and the /docs folder
- ☑ Stage 1 issues are closed on the Project board through reviewed pull requests
- ☑ The commit is tagged milestone-2-stage-1 and the instructor can open the repository

A6 Stage 1 Self-Check

- ☑ Does every core user flow have a corresponding screen?
- ☑ Do all screens share the same style (colors, fonts, buttons, spacing)?
- ☑ Does the navigation map use the exact screen names from the wireframes?
- ☑ Does every screen map to a feature in A1?
- ☑ Have we recorded every AI-assisted step in the AI Use Log (Part C)?
- ☑ Have we deleted all instruction boxes and example hints?

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PART B STAGE 2 — Architecture & Documentation

Complete this part for the Stage 2 submission only • 50 points

B1 Response to Stage 1 Feedback

Feedback received	What we changed (or why not)	Where (section / file)

B2 System Architecture

Architecture diagram

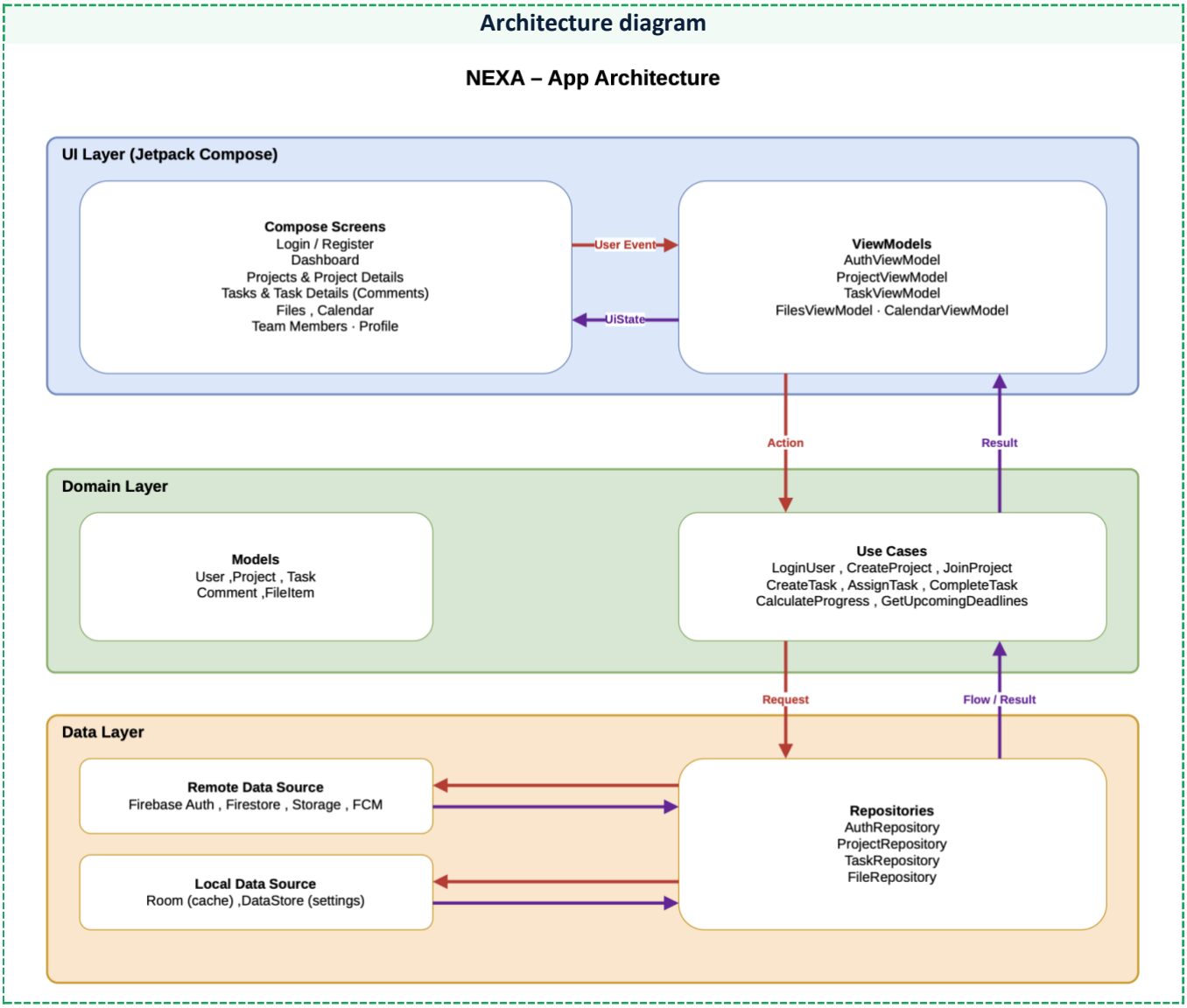


Figure 3 — NEXA app architecture: Jetpack Compose UI layer (screens and ViewModels), Domain layer (models and use cases) and Data layer (repositories with remote and local data sources). Events and actions flow down; UI state and results flow up.

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Layer responsibilities

Layer	Responsibility	Key classes	Rule
UI	Shows Jetpack Compose screens, turns user input into events and displays the UiState exposed by the ViewModels.	Compose screens (Login/Register, Dashboard, Projects & Project Details, Tasks & Task Details with comments, Files, Calendar, Team Members, Profile); AuthViewModel, ProjectViewModel, TaskViewModel, FilesViewModel, CalendarViewModel; NavHost	Must not access Firebase, Room or repositories directly; no business logic inside composables.
Domain	Holds the app's business rules as use cases and the data models they work with.	Models: User, Project, Task, Comment, FileItem. Use cases: LoginUser, CreateProject, JoinProject, CreateTask, AssignTask, CompleteTask, CalculateProgress, GetUpcomingDeadlines	Must not depend on UI classes or on Firebase / Room classes.
Data	Reads, writes and synchronises data between the remote and local data sources and gives the Domain layer a single source of data.	Repositories: AuthRepository, ProjectRepository, TaskRepository, FileRepository. Remote: Firebase Auth, Firestore, Storage, FCM. Local: Room (cache), DataStore (settings)	Must not reference ViewModels or composables; the UI never talks to data sources directly.

Unidirectional Data Flow

Example screen	New Task (screen 9) — created through TaskViewModel
UiState	Loading (task is being saved); Success (task saved → Task Created screen); Error (empty title: "Task not created" banner and inline title error, all other input kept).
Events (UI → ViewModel)	Edit title / description, choose project, assignee, due date and priority, tap "Create task", tap Cancel.
Explanation	The New Task composable only draws the current UiState and sends user actions up to TaskViewModel as events. TaskViewModel validates the input and calls the CreateTask use case, which passes the request down through TaskRepository to Firestore and the Room cache. The result travels back up the same path, and TaskViewModel publishes a new UiState (success or error). The screen observes that state and redraws, so state always flows down and events always flow up.

Walk-through of one feature

Step	Layer	What happens (use real class and function names)
1	UI	The user fills in the New Task form and taps "Create task"; the composable sends a create-task event to TaskViewModel.
2	UI (ViewModel)	TaskViewModel checks the input. If the title is empty it sets an error UiState (banner and inline message, other fields kept); otherwise it sets Loading and calls the CreateTask use case.
3	Domain	CreateTask builds a Task model (title, project, assignee, due date, priority, status To Do) and asks TaskRepository to save it.
4	Data	TaskRepository writes the task to Cloud Firestore (remote data source) and updates the Room cache (local data source).
5	Data → Domain	The repository returns the result (Flow / Result) to the use case, which passes it back to TaskViewModel.
6	UI	TaskViewModel publishes a Success UiState; the New Task screen observes it and navigates to Task Created (route task_created/{taskId}).

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B3 Jetpack Components and Libraries

Library	Used for	Why we chose it
Jetpack Compose	Build screens such as Login, Dashboard, Projects, Tasks, and Team Members.	Enables reusable UI components and updates the interface when the observed application state changes.
ViewModel	Manage screen state and handle user actions through ViewModels such as AuthViewModel, ProjectViewModel, and TaskViewModel.	Separates screen logic from composables and retains UI state across configuration changes.
Room Database	Cache project and task data locally.	Supports structured local queries and access to previously cached data without an internet connection. Repositories will coordinate updates between Room and Firestore.
DataStore	Store local user settings and preferences.	Provides asynchronous persistence for small settings, keeping user preferences available between app launches.
Firebase Authentication	Support account registration, login, logout, and password reset.	Provides managed authentication without requiring the team to develop a custom password authentication service.
Cloud Firestore	Store shared projects, tasks, memberships, deadlines, and task comments.	Supports real-time data updates so team members can observe changes to tasks and project progress.
Cloud Storage for Firebase	Store and retrieve project documents and attachments.	Handles file contents separately from structured project data and supports access control through security rules.
Firebase Cloud Messaging (FCM)	Receive notifications about task assignments, updates, and deadlines.	Enables push notification delivery from a trusted backend. Deadline scheduling and notification triggers require separate backend logic.

B4 Stage 2 GitHub Evidence

Repository structure

- ☑ README. md updated with design links and the app description
- ☑ docs/wireframes/ (one image per screen)
- ☑ docs/navigation-map.png
- ☑ docs/architecture.png and docs/architecture.md
- ☑ docs/style-guide.md
- ☑ docs/ai-use-log.md
- ☑ The commit is tagged milestone-2 and the instructor can open the repository

Project board — Milestone 2 issues

Issue	Title	Owner	Status	Pull request
[2]	*Upload architecture digram*	*MunirahHassan*	*Closed*	#PR 4 - merged
[5]	Upload navigation map	jawaher-123	closed	#PR 12 - merged
[6]	Upload and Organize Milestone 2 Design Documentation	Ghaida AlQarni	Closed	#PR 7 - merged

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Issue	Title	Owner	Status	Pull request
[7]	Document Jetpack Components and Libraries	Fatimah Alarfaj (FoFoCS)	Closed	#PR 11 - merged
[8]	*docs/wireframes/ (one image per screen)*	*HEBA ASKAR*	*Closed*	*N/A*

B5 Stage 2 Self-Check

- ☑ Does the architecture diagram clearly separate UI, Domain and Data layers?
- ☑ Is the data flow between layers shown, and explained with a real feature?
- ☑ Do the route names and screen names match the wireframes and navigation map?
- ☑ Is every library in the component plan justified?
- ☑ Are all design files committed to /docs and all Milestone 2 issues closed?
- ☑ Has every team member made at least one merged pull request?
- ☑ Have we responded to every Stage 1 feedback item?
- ☑ Have we completed the AI Use Log and deleted all instruction boxes and example hints?

PART C AI Use Log & Team Declaration

Include in both stages • add new entries in Stage 2 • keep docs/ai-use-log.md in sync

Stage / Week	Tool	Where used	Purpose, and what we did with the output
Milestone 1 (Proposal)	ChatGPT	Proposal — project research	Used to research similar project-management apps (Pair Programmer use); the team wrote its own proposal.
Milestone 1 (Proposal)	ChatGPT	Proposal — features and UI ideas	Used to get suggestions for NEXA's features and user interface (Pair Programmer use).
Stage 1	Claude	Stage 1 (Parts A–C)	Used as an assistant to format the template, refine the English phrasing of the screen descriptions, and organize the routing table structure. All core ideas, designs, and logical workflows were originally developed by the team.
Stage 2	ChatGPT	GitHub Documentation – docs/style-guide.md	Assisted in organizing and formatting NEXA's existing design guidelines, including colors, typography, spacing, and UI components, into a Markdown file for GitHub documentation. The content was based on the team's original design decisions.
Stage 2	ChatGPT	Navigation Map — A4 / docs/navigation-map.png	Used to identify missing navigation flows and generate an updated navigation map image based on the team's original map and documented screens. The generated image was added to the report and the GitHub repository.

Team declaration

- ☑ All meaningful AI assistance used in this milestone is listed in the log above.
- ☑ Every team member can explain every wireframe, diagram and decision in this document.
- ☑ The architecture and designs were created by our team, not generated wholesale by AI.

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☑ The work in this submission is our own, and each member's contribution is accurately described in the roster.

Team member	Student ID	Date
Ghaida Hashem AlQarni	2240004359	26-10-2026
Fatimah Emad Alarfaj	2240007580	26-10-2026
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